Integration Implementation

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APPENDIX A: RESTOREPOINT **14**

Overview

This document is for the services team that will deploy and configure the SL1 PowerFlow and SL1 Restorepoint products.

What is Integration Implementation?
Integration Implementation is a refined methodology for delivering the SL1 PowerFlow and SL1 Restorepoint products in the ScienceLogic cloud (SaaS). Formed to deliver a defined set of deployment goals and client expectations within a defined set of time. A project manager will be tracking the delivery to ensure all tasks are completed punctually to achieve the stated objectives.

How is this different from the previous deployment methodology?
ScienceLogic defines what the client will get, and in what timeframe, versus a more open-ended engagement where the client drives what activities they want to perform, with no explicitly defined outcomes. There are tight parameters on what's included regarding:

- PowerFlow Core Configuration
- PowerFlow Event to Incident Integration
- PowerFlow CMDB
- PowerFlow Change Management
- PowerFlow Event Management
- Restorepoint Core Configuration
- Restorepoint Integration with PowerFlow
- Restorepoint Event Enrichment

Why is Integration Implementation needed?
The SL1 product has many features, buttons, knobs, and settings. This can be overwhelming for the novice and the path to observability could be viewed as arduous. Having a playbook simplifies the one-time setup tasks and provides a prescriptive list of steps. Using the playbook will lead to:

- Rapid Deployment
- Prescriptive Delivery
- Increased Ease of Use Perception
- Repeatable, consistent deployment process
- Defined outcomes.

In support of Integration Implementation, this technical summary will detail the following:

- What is included?
- Process
- Technologies
- Tracking optics

Abbreviations and Acronyms

The following list defines the abbreviations and acronyms used in the deployment workflow:

- CLI – Command Line Interface
- CSM – Customer Success Manager
- DC – SL1 Data Collector (also called a node, appliance, or a module)
- MC – SL1 Message Collector (also called a node, appliance, or a module)
- PF - PowerFlow
- PSE – Professional Services Engineer
- PM – Project Manager
- RP – Realization Plan
- SaaS - Software as a Service
- SDD – Solution Design Document
- SL1- The ScienceLogic platform

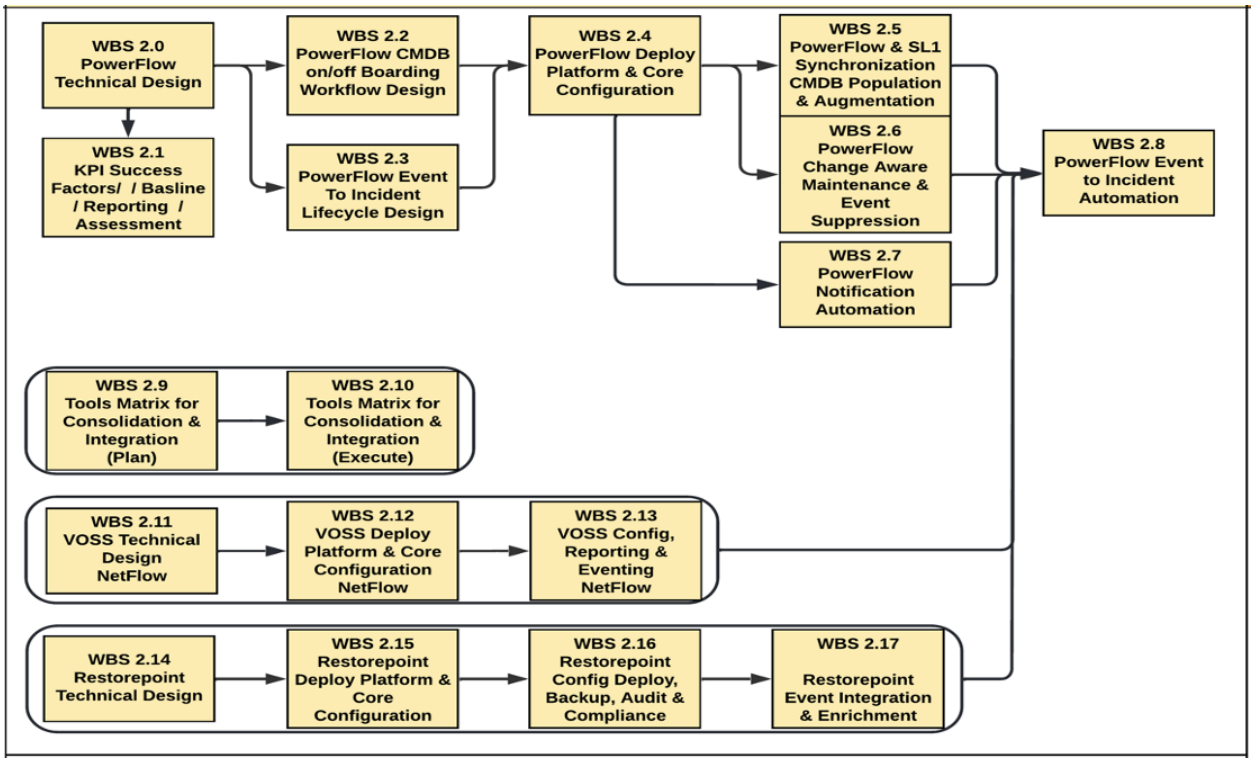
What's Included

In the Integration phase, the client will receive the following:

- SL1 PowerFlow solution to support contractual device count.
- SL1 PowerFlow to ServiceNow Integration for Incident creation (optional).
- SL1 PowerFlow to ServiceNow Integration for CMDB synchronization (optional).
- SL1 PowerFlow to ServiceNow Change Management (optional).
- SL1 PowerFlow to ServiceNow Event Management (optional).
- SL1 Restorepoint solution to support contractual device count (optional).
- SL1 best practices.
- Intelligent discovery for the contractual number of devices purchased or 5k (whichever is reached first).
- Dashboard visibility for specific technology health.
- Technology specific automations for event enrichment.
- SL1 Core Infrastructure Business Service
- Event, Device, and Log Tuning
- Self-Assessment of operational readiness.

Automation Process

The following image shows the use case process for Integration Implementation (phase 2):



Use Case 2.14 – Restorepoint Technical Design Phase

Goal: Complete the SL1 Restorepoint Design.

Steps/Tasks:

Task	Owner	Result
1. Design created as part of the foundation Solution Design Workshop. (No actions required; design is in the Solution Design Document)	Expert Services	
2. Restorepoint Design Questionnaire provided in Appendix A.	Expert Services	
3. Restorepoint Database sizing see Appendix A.	Expert Services	
4. Restorepoint Agent sizing see Appendix A.	Expert Services	
5. Review Restorepoint architecture diagram with client, see Appendix A.	PSE	
6. Review port requirements with client	PSE	
7. Review Restorepoint workflow with client	PSE	

Deliverables / Output(s):

Description	Owner	Result	Notes
1. Completed Restorepoint design (part of the Solution Design Document) based on the contract and client requirements.	Expert Services		
2. Completed Restorepoint questionnaire	Expert Services		
3. Final sizing for Restorepoint Database and agents if necessary.	Expert Services		
4. Completed Restorepoint architectural diagram	Expert Services		

Use Case 2.15 – Restorepoint Deploy Platform & Core Configuration

Goal: Validate Restorepoint platform configurations.

Steps/Tasks:

Task	Owner	Result
1. Validate Restorepoint URL and Login credentials: a) Once logged in, navigate to Information > Status page. b) Check the Appliance Software widget and verify that the License Expiry date is correct.	PSE	
2. Validate SSL certificate: a) Navigate to Administration > System Settings > Security. b) Click on “Change” Under HTTPS Certificate c) Validate the Certificate values from the pop-up window.	PSE	
3. Validate High Availability: a) Navigate to Administration > System Settings > HA. b) Verify that Member Status is Active c) Verify the Cluster Status is “Syncing”	PSE	
4. Create a SNMP credential in SL1 (ON-PREMISES ONLY) a) Create an SNMPv3 credential System>Manage>Credentials: b) Profile Name = Restorepoint v3 c) SNMP Version = SNMP V3 d) Security Name = Restorepoint e) Security Passphrase = rpdefault f) Authentication Protocol = MD5 g) Security Level = Authentication and Encryption h) Privacy Protocol = DES i) Privacy Protocol Pass Phrase = rpdefault	PSE	
5. Configure SNMP on Restorepoint Databases (ON-PREMISES ONLY) a) Navigate to Administration > System Settings > SNMP. b) Select SNMP v3 under SNMP Version c) Provide the following: ○ SNMP v3 Username = restorepoint ○ Security Level = authPriv ○ Authentication Password = rpdefault ○ Authentication Type = MD5 ○ Privacy Password = rpdefault ○ Privacy Type = DES	PSE	
6. Confirm DNS settings: a) Navigate to Administration > System Settings > Network. b) Under IP Configuration, verify that the DNS server IPs are correct.		
7. Confirm Time Server settings: a) Navigate to Administration > System Settings > Appliance. b) Under Date / Time, Verify Use NTP is checked. c) Verify the correct Time Server IPs provided.	PSE	
8. Update Customer Branding: a) Navigate to Administration > System Settings > Appliance. b) Upload customer logo image under Branding > logo > Change.	PSE	

Task	Owner	Result
9. To configure SAML: a) Navigate to Administration > System Settings > Auth Servers. b) Paste the IdP Metadata into the Identity Provider Settings. To configure LDAP: c) Navigate to Administration > System Settings > Auth Servers. d) Populate all fields.	PSE	
10. Confirm Security settings: a) Navigate to Administration > System Settings > Security. b) Under Services, verify the following options are checked: ○ Disable Inbound HTTP ○ Disable Inbound TFTP ○ Disable Inbound FTP	PSE	
11. Install Restorepoint SyncPack on PowerFlow: a) Download the latest version of the SyncPack from the ScienceLogic Support Portal by searching for "Restorepoint Synchronization". b) Login to PowerFlow. c) Navigate to SyncPacks, click Import SyncPack. d) Import the Restorepoint SyncPack. e) Activate & install the imported Sync Pack, by clicking on the hamburger menu of the imported SyncPack and clicking on Activate & Install. f) Verify that the Sync Pack has the correct Version highlighted in Green.	PSE	
12. Install Restorepoint PowerPack on SL1: a) Download the latest version of the PowerPack from the ScienceLogic Support Portal by searching for "Restorepoint". b) Login to SL1. c) Navigate to System > Manage > PowerPacks, click on Actions > Import PowerPack. d) Import and install the Restorepoint PowerPack.	PSE	
13. Install and compile Restorepoint MIBs on SL1: a) Login to the Restorepoint UI and navigate to Administration > System Settings > SNMP. b) Download the Restorepoint Appliance MIB and Restorepoint MIB. c) Login to SL1 and navigate to System > Tools > Mib Compiler and click the import button. d) Import both Restorepoint MIB (one at a time). e) Once both MIBs are imported, search for Restorepoint in the Search MIB Name. f) Select Restorepoint MIBs and select Compile MIB from the bottom right dropdown and click Go.	PSE	
14. Install Datacenter Automation Utilities PowerPack on SL1: a) Download the latest version of the PowerPack from the ScienceLogic Support Portal by searching for "Datacenter Automation". b) Login to SL1. c) Navigate to System > Manage > PowerPacks, click on Actions > Import PowerPack. d) Import the Datacenter Automation Utilities PowerPack.	PSE	

Task	Owner	Result
15. Create SSH Credential in SL1 for the Network devices. a) Navigate to System > Manage > Credentials. b) Click Actions > Create SSH/Key Credential c) Provide a name for the credential. d) Set Hostname = %D e) Set Port = 22 f) Set timeout = 2000 g) Set username to client provided username to ssh into the devices. h) Set password to client provided password to ssh into the devices. i) Save and exit j) If multiple ssh credentials need to be created, then follow the above steps to create more credentials.	PSE	
16. Get Restorepoint API Token a) Navigate to Administration > API Tokens. b) If a token already exists, then copy and save the token value; this will be used in the next step. c) If a token does not exist, then click on Add Token: ○ Provide a description and click Submit. ○ Copy and save the token value for the new token.	PSE	
17. Create Configuration Object in PowerFlow. a) Login to PowerFlow b) Navigate to Configurations. c) Edit Configuration named Restorepoint Base Config, by clicking on the Actions icon and clicking Edit. d) Click Copy as e) Complete the following fields: ○ Friendly Name: Name of the configuration object that will display on the Configurations page. ○ Description: A brief description of the configuration object. ○ Author: User or organization that created the configuration object. ○ Version: Version of the configuration object. c) In the Configuration Data field, update the default variable definitions to match the PowerFlow configuration: ○ sl1_url: Type the URL for your associated SL1 system. ○ sl1_user: Type the username for your SL1 system. ○ sl1_password: Type the password for your SL1 system. ○ sl1_db_host: Type the URL for your associated SL1 database. ○ sl1_db_user: Type the username for your SL1 database. ○ sl1_db_password: Type the password for your SL1 database. ○ restorepoint_url: Type the URL for the Restorepoint system. ○ restorepoint_api_token: Type the API token for your Restorepoint system, from previous step. ○ default_restorepoint_device_type: Type the default device type for the Restorepoint system, out of the box value is “a10_agalaxy”. ○ default_backup_interval: Type the default time for the Backup Interval for the Restorepoint device. Customers	PSE	

Task	Owner	Result
will provide the desired default configuration backup interval for their devices. ○ create_custom_link: Type a value to create an optional custom link from SL1 to Restorepoint. This field is optional. ○ restorepoint_ui_url: Type the URL for the Restorepoint system. ○ Restorepoint_config: Type Enable or Disable to allow device change detection. ○ Timestamp: Use Default. ○ default_monitoring_monitor_device: Type True or False to enable or disable device monitoring. The default value is True. ○ default_monitoring_fail_after: Type how many failed attempts to onboard a device before PowerFlow will stop attempting to discover the device. ○ default_monitoring_is_ping_type: Type True or False to enable or disable ICMP ping rather than TCP connection. The default value is True. ○ default_monitoring_email_when_down: Type True or False to enable or disable sending an email when the device is down. The default value is False. ○ default_monitoring_email_when_up: Type True or False to enable or disable sending an email when the device is back up. The default value is True. d) Click Save.		
18. Configure the PowerFlow Applications: a) Login to PowerFlow and navigate to Applications. b) Search for Restorepoint: Sync Devices under Application Name. c) Click on the Application name. d) Click on Configure (top right). e) Under configuration, select the new configuration that was created in the previous step. f) Click Save and exit. g) Navigate to Applications. h) Search for Restorepoint: Sync Devices under Application Name. i) Click on Schedule. j) Provide a Schedule Name. k) Set Frequency value to 900. l) Click Save Schedule. m) Follow the above steps a – l for “Restorepoint: Get the list Of Logs from Restorepoint” application	PSE	
19. Verify Device Sync is successful. a) Login to PowerFlow and navigate to Applications. b) Search for “Restorepoint: Sync Devices” under Application Name. c) Click on the Application name. d) Click on Run. e) Verify that all steps are showing Green. f) Login to Restorepoint. g) Navigate to Devices > Device List h) Verify that devices are being created in Restorepoint.	PSE	

Task	Owner	Result
i) Pick one newly created device, copy the name, IP, and Domain of the device. j) Login to SL1. k) Navigate to Devices. l) Search for the Device Name and IP copied from Restorepoint in the previous step. m) Click on the device and open the Device Investigator. n) Click on the Attributes tab. o) Verify that the Restorepoint ID attribute is populated with the device ID from Restorepoint.		
20. Discover in-scope devices not currently in SL1 and / or confirm the Restorepoint Devices Device Group is populated with the in-scope devices.	PSE	
21. Verify Custom Link creation on the SL1 Devices.		
22. Knowledge transfer and transition administration to client.	PSE	

Deliverables / Output(s):

Description	Owner	Result	Notes
1. Completed initial Restorepoint platform configuration.	PSE		
2. Restorepoint system settings are configured.	PSE		
3. Restorepoint High Availability configured.	PSE		
4. Restorepoint Authentication configured.	PSE		
5. Restorepoint integration to SL1 configured and tested.	PSE		
6. Confirm the devices from the Restorepoint Devices Device Group have synced to the Restorepoint system, and they have the Restorepoint ID attribute populated with the corresponding Restorepoint Device ID.	PSE		

Use Case 2.16 – Restorepoint Config Deploy, Backup, Audit & Compliance

Goal: Restorepoint backup, restore, audit, and compliance.

Steps/Tasks:

Task	Owner	Result
1. Identify a test group of devices to perform backups.	PSE	
2. Run a backup: a) Navigate to Devices > Device List. b) Select the checkboxes for the devices identified in step 1. c) Click the Backup button. d) Monitor the running tasks at the bottom of the screen by clicking on the Tasks link at the bottom of the page.	PSE	
3. Validate successful backup of network devices in Restorepoint. a) Navigate to Devices > Device List. b) Click on one device from the list of devices identified in step 1. c) Navigate to Configurations. d) Sort by Date to get the newest backup on the top. e) You should see the latest backup details in the file.	PSE	
4. Perform test restore of configuration: a) Navigate to Devices > Device List. b) Select a device from the list of devices identified in step 1. c) Navigate to Configurations. d) Select the latest backup file from the list. e) Click Restore. f) Click Save.	PSE	
5. Perform test restore of configuration: g) Navigate to Devices > Device List. h) Select a device from the list of devices identified in step 1. i) Navigate to Configurations. j) Select the latest backup file from the list. k) Click Restore.	PSE	

Deliverables / Output(s):

Description	Owner	Result	Notes
1.			
2.			
3.			
4.			

Use Case 2.17 – Restorepoint Event Enrichment

Goal: Setup Event Enrichment for Restorepoint related events.

Steps/Tasks:

Task	Owner	Result
1. Install Restorepoint Automation PowerPack on SL1: a) Download the latest version of the PowerPack from the ScienceLogic Support Portal by searching for “Restorepoint Automation”. b) Login to SL1. c) Navigate to System > Manage > PowerPacks, click on Actions > Import PowerPack.	PSE	

2. Create Restorepoint Credential in SL1.
a) Navigate to System > Manage > Credentials.
b) Search for Restorepoint Automation and edit the credential.
c) Update the name of the credential to something new.
d) In the credential, provide the IP, username and Password for the Restorepoint System.
e) Click Save As.
f) This will create a new credential, save the Credential ID of the newly created credential.
g) Navigate to Registry > Run Book > Actions.
h) Search for Restorepoint.
i) Edit the first Action.
j) In the Input Parameters section, save the Credential ID from above into the `"sl1_credential_id"` value, the code should look something like:
`{`
`"sl1_credential_id":"62",`
`"max_log":"",`
`"action":"recent_backups_diff"`
`}`
k) Save nad exit
l) Make the same update to the remaing actions in the list from step h.
	PSE	

3. Verify that the Restorepoint Automations are enabled: a) Navigate to Registry > Run Book > Automation. b) Search for Restorepoint. c) Verify that all Restorepoint policies are enabled. If they are not enabled, then enable them.	PSE	

Deliverables / Output(s):

Description	Owner	Result	Notes
1.			
2.			
3.			
4.			

Appendix A: Restorepoint

Restorepoint Design Questionnaire

Question	Answer
1. Will the SL1 Restorepoint solution be hosted in ScienceLogic SaaS or on-premises?	
2. If not in SaaS, what deployment platform: • VMware (ESX vSphere 6.7 Update 2 or higher) • Restorepoint Appliance	
3. Was High-Availability purchased for Restorepoint?	
4. Where will the Restorepoint Database be located?	
5. On what site will the Restorepoint HA Database be located?	
6. How many device licenses were purchased for Restorepoint?	
7. Based on the number of devices, what is the recommended size of the SL1 Restorepoint database?	
8. Will SL1 Restorepoint Agents be part of the architecture?	
9. If agents are being used, which sites will they be deployed to?	
10. Are High-Availability agents required?	
11. Validate if CA SSL certificate is needed?	
12. What authentication method will be used? a. Radius b. SAML c. LDAP	If SAML, provide IdP Metadata If LDAP, see Restorepoint UI>Administration>Auth Servers>LDAP

Notes:

Restorepoint SSL

Restorepoint initially uses a self-signed certificate. Because of this, your web browser will warn you of an invalid (untrusted) certificate. This is normal behavior because the appliance certificate is not signed by a Trusted Certificate Authority. The session will still be encrypted. Refer to your browser for instructions on how to proceed and accept the unsigned certificate. A valid (signed) certificate can be uploaded to Restorepoint after the initial configuration is completed.

Encryption

All sensitive data stored in Restorepoint, including device configurations, is protected by encryption. Restorepoint encrypts data when it is written to a disk and decrypts it as it is read. Cleartext data is only held in volatile memory. Therefore, the data disappears when the appliance is shut down or rebooted, which renders data theft impossible
without a valid encryption key.

Restorepoint Sizing

Appliance	CPU	Memory	OS Disk	Data Disk
Master DB	1	8 GB	20GB	256GB
Secondary DB	1	8 GB	20GB	265GB

Appliance	CPU	Memory	OS Disk	Data Disk
Agent	1	4GB - minimum 8GB - recommended	20GB	20GB

Restorepoint Database Firewall Requirements

Clients to Restorepoint

Port	Purpose
TCP/443	Restorepoint UI/API
TCP/22	Restorepoint Shell access
UDP/161	(optional) SNMP monitoring of Restorepoint DB

Traffic from Restorepoint to Network Devices

Port	Purpose
TCP/22 or device specific ports from Restorepoint Plugin Guide.	Perform configuration backups

Other traffic originating from Restorepoint

Port	Purpose
TCP/443	Download updates from Restorepoint update servers, and HA database sync
UDP/53	Lookups to DNS servers
TCP/25	Send notification emails using SMTP
UDP/123	Time synchronization with NTP servers (optional)
TCP/22	Initiate remote support requests (jmp1.restorepoint.com and jmp2.restorepoint.com), or communicate with an Agent's master (optional)

PowerFlow to Restorepoint

Port	Purpose
TCP/443	Restorepoint UI/API

Restorepoint Agents

Agents allow a Restorepoint appliance to manage devices located on a remote or otherwise disjoint network, not directly routable by Restorepoint, without complex firewall changes, Network Address Translation, or VPNs. For instance, a Service Provider can set up a central Restorepoint appliance and deploy agents on customer networks and enable device backups on remote sites.

An Agent can be deployed as a Virtual or Hardware appliance on the remote network. The agent provides fast operations by locally performing all the tasks that would typically require extensive network interaction. Configurations, logs, etc. are processed locally by the agent, and uploaded to the Master Restorepoint appliance.

Restorepoint Agents Firewall Requirements

An agent initiates and maintains an SSH connection to the master Restorepoint appliance to receive tasks to execute, upload and download device configurations, task output and logs, and download software updates.

Your firewall policy must allow SSH traffic (TCP port 22) from the agent to the master for an agent to function correctly.

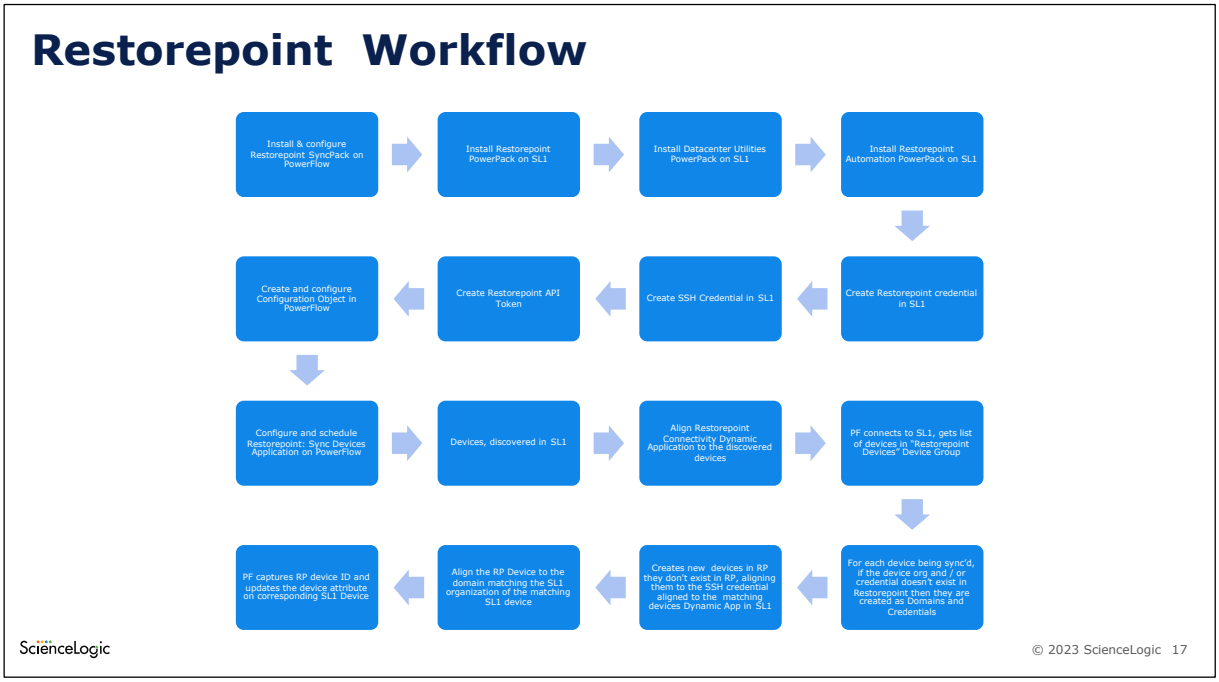
Agents to Restorepoint Master DB

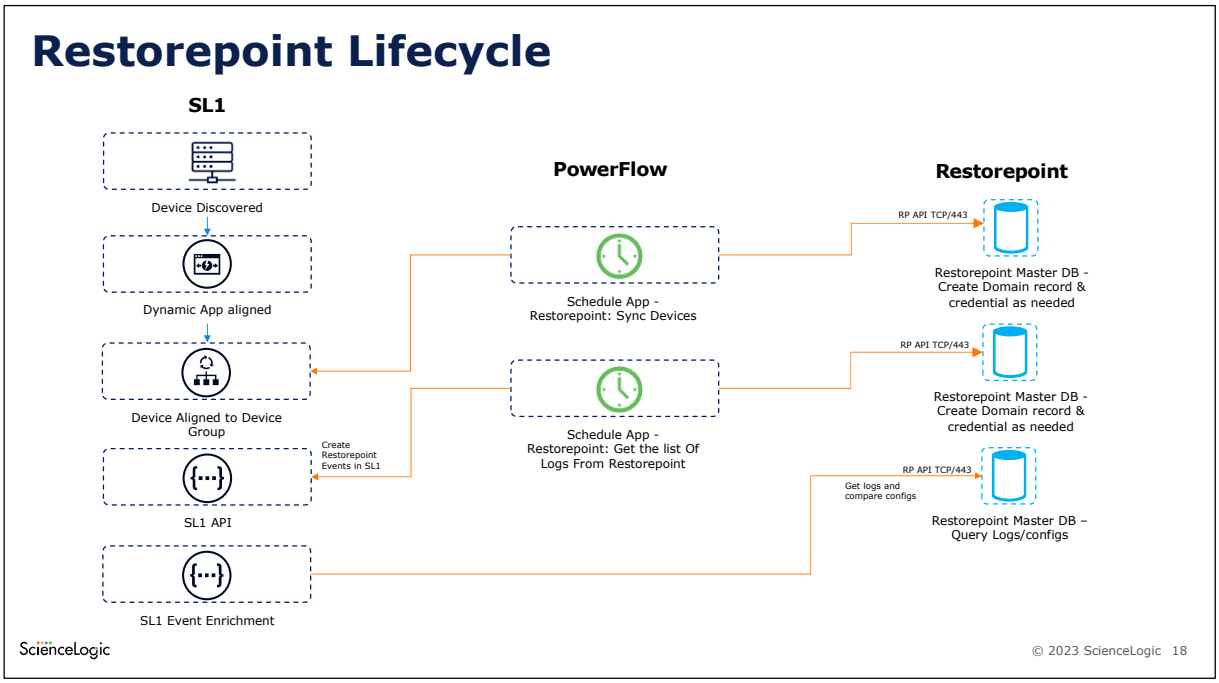
Port	Purpose
TCP/22	SSH connection to Restorepoint Master and Secondary DB

Restorepoint High Availability

High Availability (HA) provides a way to minimize the effects of hardware failure by configuring two Restorepoint appliances in a cluster. Under normal operating conditions, the primary cluster member is active, and the secondary is in standby mode; the active appliance performs all network operations and replicates all settings and device configurations to the standby appliance. Restorepoint replicates data both incrementally (for example, just after a backup is retrieved from a device) and by performing full synchronizations on a regular basis. If the primary member becomes unavailable because of a hardware failure, other network problems, or from losing power, the secondary member will automatically become Active and carry on as normal. If the primary recovers, it will automatically take over from the secondary and become active. HA does not require the appliance to be installed on the same network, if the traffic requirements are met (see below). Software updates and upgrades are managed at the cluster level; updating the active appliance will automatically update the standby appliance.
HA is a separately licensed feature. Only appliances of the same model can be clustered, and appliances must be running the same software
version.
Cluster members must be able to communicate over HTTPS to exchange heartbeat information and data synchronization. TCP/443 traffic should be permitted bidirectionally between the appliances.

Restorepoint Diagrams





Restorepoint PowerPacks

PowerPack Name	Notes

1. Restorepoint PowerPack	**Contents:**
	• 1 x Device Class: Restorepoint
	• 1 x Dynamic Application: Restorepoint Connectivity
	• 14 x Events:
	  ○ 10 x Trap Event Policies
	  ○ 2 x Syslog Event Policies
	  ○ 2 x API Event Policies

2. Restorepoint Automation PowerPack	
3. Restorepoint Synchronization SyncPack*	Requires Subscription to Advanced or Premium License